Misallocated, Not Mispriced: The Resolution Limit of Local Strength in the Factor Zoo

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Abstract

The factor zoo throws away live factors, and I show exactly how many, why, and how far they can be reclaimed. A universal significance hurdle keeps a factor only if its t-statistic clears a threshold in the broad market. The rule is mis-specified, not merely conservative: it judges every factor in a single venue when strength is local to another. Of 179 anomalies, the 94 that a $|t| < 1.96$ screen rejects in large-capitalization stocks earn an out-of-sample Sharpe ratio of 0.85 where they are tested, yet 1.30 when reallocated as a group to the market segment they fit—a six-factor alpha of 3.8% per year ($t = 8.3$) that survives q-factor spanning, trading costs, and half a century of data never used in selection. I explain both the gain and its ceiling with a single phase transition. Casting factor-to-venue assignment as a Roy self-selection problem, I show that recovering a factor's comparative advantage is a spiked-covariance detection problem governed by a Baik–Ben Arous–Péché threshold: identifiability depends on a signal-to-noise ratio that scales with the number of factors pooled into the estimate. Coarse reallocation of the discard pool sits far above the threshold and is almost perfectly recoverable; factor-by-factor routing sits below it and is information-theoretically impossible. This makes the blunt coarse move not a heuristic but a capacity: across five routers chosen to beat it—per-factor argmax, economic-family priors, trailing-performance timing, a machine-learning characteristic map, and an empirical-Bayes shrinkage rule, each tuned out of sample—not one reliably improves on coarse by more than noise, because only six of ninety-four factors clear the per-factor threshold and

*Replication archive: `https://nemo02070118.github.io/boatfx-repro/`. I thank, without implicating, seminar participants for comments. All errors are my own.

the rest are locked by the data-processing inequality. The lesson for a literature chasing conditional factor timing is exact: the simple reallocation already extracts what a finite sample permits anyone to extract. The welfare cost of the hurdle grows with the zoo: as the candidate count rises, the discovery threshold diverges from the allocation threshold, and the mass of wrongly discarded local strength expands without bound.

Keywords: factor zoo, multiple testing, comparative advantage, random matrix theory, phase transition, limits to arbitrage.
JEL classification: G11, G12, G14, C58.

Introduction

Consider an anomaly whose long–short return, measured across all stocks, carries a t-statistic of 1.4. Standard practice retires it. It fails the 1.96 hurdle that the factor-zoo literature erects to keep false discoveries out of a crowded field, so it is deemed spurious and set aside. This paper shows that the verdict is frequently a mistake about *venue* rather than *value*. The same factor, traded in the slice of the market it is built to exploit instead of the broad capitalization-weighted universe where convention evaluates it, earns economically large and statistically decisive returns out of sample.

The facts are stark. I study 179 anomalies from the open-source cross-section of Chen and Zimmermann [2022], each realized in four nested equity universes that differ in the size and liquidity of their constituents: capitalization-weighted large caps, NYSE-breakpoint portfolios, a size-20 micro-cap construction, and equal-weighted small caps. I split the 1926–2024 sample in half. On the first half I apply the textbook screen, labeling a factor *discarded* if its large-cap t-statistic falls below 1.96; 94 of the 179 factors fail. I then hold these 94 discards over the second half, 1975–2024—594 months never touched in selection—and compare two portfolios. Left in large caps, where the profession rejected them, the equal-weighted discard book earns an annualized Sharpe ratio of 0.85. Reassigned as a single pool to the small-cap segment that their first-half record identifies as their home, the *same* factors earn 1.30. The improvement of 0.45 Sharpe units has a bootstrap 95% interval of $[0.25, 0.67]$ and is positive in 1.000 of 5,000 resamples. Three-quarters of the way to the unattainable hindsight ceiling of 1.45, the gain is recovered by a rule a practitioner could have run in 1975.

This is not a size premium in disguise. Against the Fama–French six-factor model the reallocated book earns an alpha of 3.8% per year with a Newey–West t of 8.3; against the Hou–Xue–Zhang q-factor model, 3.4% with $t = 7.5$. The *incremental* return from reallocation—the long–short of the reallocated book against the same factors left in place—is itself an alpha of 2.3% per year ($t = 5.1$) and survives spanning by the surviving canonical factors ($t = 4.3$). A joint test that the family-level reallocation gains are all zero is rejected at $F(8, 594) = 5.7$, $p < 0.0001$. The gain holds under a twelve-month embargo between estimation and trading, under expanding-window walk-forward re-estimation ($t = 5.6$), in eight of nine decades including the 1930s, and after transaction costs up to roughly 85 basis points per one-way trade. For a mean-variance investor it is worth about 1.6% per year in certainty equivalent, essentially independent of risk aversion.

Why does a universal hurdle discard value, and—just as important—why has the profession not already recovered it with a smarter, factor-by-factor router? Both questions have one answer, and it is the theoretical core of the paper. I cast the assignment of factors to

venues as a Roy (1951) self-selection problem: each factor has a venue-specific productivity, and an omniscient planner would route it to the venue of its comparative advantage. The analyst, however, sees only a noisy, finite-sample estimate of that advantage. Whether the true ranking of advantages can be recovered from the estimate is exactly a spiked-covariance detection problem, governed by the Baik–Ben Arous–Péché (BBP) threshold [Baik et al., 2005, Paul, 2007]. The recoverable overlap between the estimated and true ranking is zero below a critical signal-to-noise ratio and rises continuously above it. The signal-to-noise ratio scales with the number of factors m aggregated into the estimation unit, $\tau(m) = m\,\mathrm{Var}(a)/\sigma^2$, so *resolution*—how finely one attempts to route—determines identifiability.

Calibrated to the data, the critical unit size is $m^\star \approx 1.3$ factors, and this single number organizes every result. A *coarse* rule that moves the entire pool of 94 discards as one unit operates at $m = 94$, hence $\tau \approx 73$ and a recoverable overlap of 0.99: the reassignment is almost perfectly identifiable, and it captures 76% of the hindsight-oracle gain. A *fine* rule that routes each factor on its own operates at $m = 1$, hence $\tau \approx 0.78 < 1$ and a recoverable overlap of zero: factor-by-factor routing is information-theoretically impossible out of sample. The data confirm the prediction to the decimal. The rank persistence of a factor's own best-venue advantage, estimation half to trading half, is +0.08—noise, as the theory demands—and every attempt to beat the coarse rule with a fine one fails. An adaptive router that times venues on trailing performance scores 1.25; a machine-learning router that predicts each factor's venue from its observable characteristics scores 1.12; both fall short of the static coarse 1.30. The wall binds timing and machine learning alike because it is a property of the information, not of the algorithm.

This failure is not a verdict on the three routers I happen to try. It is a capacity statement, and it is the paper's sharpest claim against current practice. A literature now spends great effort conditioning factor exposures on states, characteristics, and machine-learned signals, in the belief that a cleverer rule will route factors to where they pay. I test five such rules—per-factor argmax, economic-family priors, trailing-performance timing, a machine-learning characteristic map, and an empirical-Bayes shrinkage rule, each tuned on a split internal to the training sample and then frozen—and none reliably beats the one-line coarse move by more than noise. The reason is exact: only six of the ninety-four discarded factors carry an own-versus-coarse signal strong enough to clear the per-factor threshold, so for the other eighty-eight the data-processing inequality guarantees that any rule is trading on noise. The coarse reallocation is therefore not a convenient heuristic that something better will displace. It already extracts what a finite sample permits anyone to extract. The contribution is a limit with teeth: it tells the conditional-timing literature not that it has not yet found the right model, but that beyond a reallocation a practitioner could run in 1975, there is no

model to find.

The final result is a comparative static with uncomfortable implications for how factors are discovered. The threshold that governs whether a factor is *discovered*—the maximum of many t-statistics—grows like $\sqrt{2 \log N}$ in the number of candidates N. The threshold that governs whether its venue can be *allocated* grows only like $\sqrt{2 \log \log N}$. The two diverge. As factor factories enlarge N by orders of magnitude, the hurdle rises much faster than the information needed to place survivors, so the pool of locally strong factors that are discarded but reallocatable expands without bound. The problem this paper identifies does not shrink as the zoo is cleaned up; it grows.

The argument speaks to three literatures and departs from each in a specific way. The first is the factor-zoo and multiple-testing literature [Harvey et al., 2016, McLean and Pontiff, 2016, Chordia et al., 2020], which raises the hurdle to protect against data mining as the count of candidates explodes. I take the hurdle as given and ask what it discards; my contribution is to show that a fixed universal threshold is mis-specified, not merely strict, because it conflates where a factor is weak with whether it is weak. The second is the literature on where anomalies live, which documents that they are stronger among small, illiquid stocks [Fama and French, 2008, Israel and Moskowitz, 2013]. My point is not that small-cap anomalies are strong—that is known—but that the factors the profession has *already rejected* harbor recoverable value, and that recovering it is a provable boundary problem with a sharp ceiling. The third is the use of random matrix theory in finance [Baik et al., 2005, Paul, 2007, Benaych-Georges and Nadakuditi, 2011, Bun et al., 2017], which cleans covariance matrices; I bring the BBP threshold to bear on factor *allocation* and derive a resolution limit that predicts both the recoverable gain and its ceiling from three measurable primitives.

The paper proceeds as follows. Section 1 develops the Roy model of factor-to-venue sorting and states the four results that organize the evidence: Proposition 1 (the universal hurdle is mis-specified), Theorem 1 (the resolution limit), Theorem 2 (discovery and allocation thresholds diverge), and Theorem 3 (the capacity of local-strength recovery and its converse), together with the timing-and-ML corollary. Section 2 describes the data and the four venues. Section 3 reports the main reallocation result. Section 4 subjects it to risk adjustment, costs, embargoes, and walk-forward re-estimation. Section 5 documents the resolution wall and the failure of fine routing. Section 6 proves the converse: a tournament of five routers, each tuned out of sample, shows none beats the coarse move by more than noise, so the simple reallocation is the capacity of the problem. Section 7 calibrates the phase transition and shows it reproduces the evidence. Section 8 develops the divergence result and its implications for the growing zoo. Section 9 concludes. Proofs, Monte Carlo

validation, data construction, and secondary robustness tests are in the Internet Appendix.

1 A Model of Factor Comparative Advantage

The empirical results below are easy to state and, I will argue, hard to dismiss. But a table of Sharpe ratios does not by itself explain why a universal hurdle discards value, nor why a cleverer rule cannot recover more of it than a blunt one. This section builds the model that answers both questions. It has three moving parts: a Roy (1951) description of where factors are strong, a decision problem for an analyst who must infer that strength from finite data, and a random-matrix characterization of when the inference is possible. The results that follow—one about the hurdle, one about resolution, one about scale, and one about the capacity of recovery and its converse—are the spine of the paper.

1.1 Setup: factors, venues, and comparative advantage

There are N candidate factors indexed by i and V venues indexed by v. A venue is a tradable universe—a capitalization segment, an exchange, a weighting scheme—in which any factor can be formed. Factor i traded in venue v delivers an excess long–short return with population mean μ_{iv} and variance normalized to one, so that μ_{iv} is the factor's *true Sharpe ratio* in that venue. Collect these in a matrix $M = [\mu_{iv}]$.

I decompose each entry into a part that is common across venues and a part that is local to a venue:

$$\mu_{iv} = \beta_i + a_{iv}, \qquad \sum_v a_{iv} = 0, \tag{1}$$

where $\beta_i = \frac{1}{V}\sum_v \mu_{iv}$ is factor i's *global strength*—how well it works on average across venues—and a_{iv} is its *local strength* in venue v, the comparative advantage that the Roy model places at center stage. By construction local strength is a pure tilt: it reallocates a factor's performance across venues without changing its average. The two objects answer different questions. Global strength β_i asks *whether* a factor works; local strength a_{iv} asks *where* it works best.

1.2 The hurdle is a test of the wrong quantity

The factor-zoo literature keeps a factor if it clears a significance hurdle in a reference venue, almost always the broad capitalization-weighted market. Write the reference venue as v_0 and the rule as: retain i if $\mu_{i,v_0} \geq h$ for a hurdle $h > 0$ (in finite samples, if its t-statistic

in v_0 clears a critical value; the population statement isolates the economics). The following observation, immediate from (1), is the conceptual pivot of the paper.

Proposition 1 (The universal hurdle is mis-specified)**.** *Suppose factor i is retained if and only if $\mu_{i,v_0} \geq h$. Then there exist factors with $\mu_{i,v_0} < h$—discarded—yet $\max_v \mu_{iv} \geq h$, so that the factor is genuinely profitable in some venue. The discarded set contains such factors whenever some factor's local strength satisfies $a_{i,v_i^\star} - a_{i,v_0} \geq h - \mu_{i,v_0} > 0$, where $v_i^\star = \arg\max_v \mu_{iv}$. A rule that evaluates every factor in the single venue v_0 therefore commits a type of error that no choice of h can remove: it rejects local strength as if it were absence of strength.*

The proof is one line and appears in Internet Appendix IA.1; the content is in the reading. The hurdle is not too high or too low. It is pointed at the wrong quantity. It measures μ_{i,v_0}, a single column of M, and discards on that basis, when the economically relevant quantity is $\max_v \mu_{iv}$, the row maximum. Raising h to control false discoveries, the response of the multiple-testing literature, makes this error *worse*, not better, because it discards more rows on the strength of one column. The question is then whether the row maximum, and the venue that attains it, can be recovered from data. That is a problem of inference, and it is where the next two results bind.

1.3 The analyst's problem and the signal-to-noise ratio

An omniscient planner who knew M would route each factor to its best venue $v_i^\star$ and capture $\frac{1}{N}\sum_i \max_v \mu_{iv}$, the hindsight oracle. The analyst does not know M. From a sample of length T she forms estimates $\hat{\mu}_{iv} = \mu_{iv} + \varepsilon_{iv}$, where the sampling error of a Sharpe ratio has variance

$$\sigma^2 = \mathrm{Var}(\varepsilon_{iv}) = \frac{1 + \frac{1}{2}\mathrm{SR}^2}{T_{\text{eff}}}, \tag{2}$$

the Lo [2002] standard error, with T_{eff} the effective number of independent observations after accounting for serial dependence. She must decide, for a chosen *unit of aggregation* of m factors, which venue the unit prefers, using only the local-strength estimates $\hat{a}_{iv}$. The unit may be a single factor ($m = 1$, fine routing) or the entire discard pool ($m = N$, coarse reallocation); resolution is the analyst's choice, and it is the choice that decides everything.

Stack the local-strength estimates for a unit into a venue-by-factor array and ask the natural question: does the leading direction of the estimated array line up with the leading direction of the true array $[a_{iv}]$? This is a signal (the true local-strength tilt) observed through additive noise (sampling error), and whether the signal survives is governed by their

ratio. Define the *local-strength signal-to-noise ratio* of a unit of m factors as

$$\tau(m) = \frac{m \operatorname{Var}_{\text{cross}}(a)}{\sigma^2}, \qquad c = \frac{V-1}{T_{\text{eff}}}, \tag{3}$$

where $\operatorname{Var}_{\text{cross}}(a)$ is the cross-venue dispersion of true local strength—how much comparative advantage there is to find—and c is the aspect ratio of the estimation problem. The quantity τ rises with the amount of real signal and the number of factors pooled, and falls with sampling noise. Everything hinges on how large it is relative to one.

1.4 The resolution limit

The estimated local-strength array is a low-rank signal buried in a noise matrix, exactly the object studied by random matrix theory. Its leading sample eigenvector—the venue tilt the analyst would act on—aligns with the truth only once the signal clears a sharp threshold. This is the Baik–Ben Arous–Péché phase transition [Baik et al., 2005, Paul, 2007], and it delivers the paper's central limit on how finely factors can be routed.

Theorem 1 (Resolution limit of local strength)**.** *Let a unit of m factors have signal-to-noise ratio $\tau(m)$ and aspect ratio c as in* (3)*. As $T_{\text{eff}}, m \to \infty$ with c fixed, the squared overlap q^2 between the estimated best-venue direction and the true best-venue direction converges almost surely to*

$$q^2(\tau) = \begin{cases} 0, & \tau \le 1, \\ \dfrac{1 - 1/\tau}{1 + c/\tau}, & \tau > 1. \end{cases} \tag{4}$$

Consequently there is a critical unit size $m^\star$, solving $\tau(m^\star) = 1$, below which the analyst's estimated routing is asymptotically uncorrelated with the truth, and above which the recoverable fraction of the oracle's local-strength gain equals $q^2(\tau(m))$ and rises continuously toward one.

The proof maps the problem onto a rank-one spiked covariance model and invokes the BBP almost-sure limit; it is given in full in Internet Appendix IA.1, with the finite-sample accuracy of (4) verified by Monte Carlo in Appendix IA.2. The economics of (4) are worth stating plainly. Below the threshold the top sample eigenvector is pure noise: it points in a direction unrelated to true comparative advantage, so acting on it adds nothing in expectation. There is no smooth decay into this region and no partial credit—the overlap is exactly zero, not merely small. Above the threshold the signal condenses, the eigenvector rotates toward the truth, and the recoverable gain grows. The transition is a genuine discontinuity in *behavior*,

and it converts the vague intuition "you need enough data" into a falsifiable inequality, $\tau(m) \gtrless 1$, whose two sides the data will occupy cleanly.

Because τ scales linearly in m, Theorem 1 is a statement about resolution. The same primitives can place a unit on either side of the wall depending only on how many factors are pooled into it. Coarse reallocation of the whole discard pool makes m large and lands far above the threshold; routing one factor at a time makes $m = 1$ and, as the calibration in Section 7 shows, lands below it. A single mechanism thus predicts both that coarse reallocation works and that fine routing cannot—not for want of a better method, but because the information to route finely does not exist in the sample.

1.5 Discovery and allocation thresholds diverge

The last result turns from a fixed zoo to a growing one and asks how the problem scales. Two thresholds govern a factor's fate. To be *discovered* it must clear a multiple-testing hurdle: with N candidates and independent noise, the largest spurious t-statistic is of order $\sqrt{2\log N}$, so the discovery threshold is $\kappa_D(N) \asymp \sqrt{2\log N}$—the familiar reason the hurdle must rise with the zoo [Harvey et al., 2016]. To have its venue *allocated*, by contrast, the analyst needs only to preserve the *ranking* of a factor's venue advantages, not to certify any single one against the global null. The relevant control is on the maximum of the estimation errors around an already-identified ordering, which concentrates at rate $\sqrt{2\log\log N}$.

Theorem 2 (Divergence of discovery and allocation)**.** *Let $\kappa_D(N) \asymp \sqrt{2\log N}$ be the discovery threshold and $\kappa_A(N) \asymp \sqrt{2\log\log N}$ the allocation threshold for a zoo of N candidate factors. Then the gap*

$$\Delta(N) = \kappa_D(N) - \kappa_A(N) \asymp \sqrt{2\log N} - \sqrt{2\log\log N} \to \infty \tag{5}$$

diverges as $N \to \infty$. Hence the measure of factors that fail discovery yet retain allocatable local strength—the discards this paper reallocates—grows without bound in the size of the zoo.

The two thresholds grow at different speeds because they answer different questions, and the iterated logarithm is the fingerprint of the easier one. Discovery fights the whole family of candidates at once and pays the $\sqrt{2\log N}$ price of the maximum; allocation only has to keep a within-factor ordering straight and pays the far gentler $\sqrt{2\log\log N}$. The practical reading is uncomfortable. Each new generation of factor factories enlarges N by orders of magnitude and, to hold false discoveries in check, raises the hurdle in step with $\sqrt{2\log N}$. But the information needed to place the survivors grows only like $\sqrt{2\log\log N}$. The scissors open. The very act of policing the zoo more strictly discards an ever larger reservoir of local

strength that remains, by Theorem 1, recoverable in aggregate. The problem is not a relic of a messy early literature that careful work will tidy away; it is a feature that intensifies precisely as the literature grows more careful.

1.6 Why timing and machine learning cannot help

A natural objection is that the fine-routing failure reflects a weak method, and that conditioning on more information—trailing performance, firm characteristics, a flexible machine-learning map—would recover the per-factor venue. Theorem 1 says otherwise, and the logic is a corollary rather than a new result.

Corollary 1 (No conditioning rescues sub-threshold routing)**.** *Any venue-routing rule for a unit of m factors is a measurable function of the in-sample estimates $\hat{a}_{iv}$. If $\tau(m) \leq 1$, the leading estimated direction carries no asymptotic information about the true best venue ($q^2 = 0$), so no measurable function of the estimates—linear timing rule, nonlinear machine-learning map, or otherwise—can achieve positive expected overlap with the oracle routing. Conditioning on auxiliary signals helps only to the extent that those signals raise τ above one; absent that, the data processing inequality caps the achievable gain at the coarse-unit value.*

The corollary draws a bright line between the two halves of the evidence. Coarse reallocation works because it pools m large enough to clear the threshold, so there is signal for *any* method to use. Fine routing fails because at $m = 1$ there is no signal, and the data-processing inequality forbids manufacturing it: a cleverer function of noise is still noise. This is why, in Section 5, an adaptive timing router and a machine-learning characteristic router both underperform the one-line static rule. They are not badly tuned. They are solving a problem the sample does not contain the information to solve. The wall is informational, and it binds every method equally.

1.7 The coarse move is a capacity

The corollary forbids per-factor routing. The next result turns that prohibition into a positive statement about the coarse move: it is not merely a good rule, it is the most any causal rule can achieve. The intuition is that the oracle gain is sparse across factors—most discards belong at the coarse venue, and only a few carry an own-versus-coarse advantage worth acting on—so the expected gain of *any* router is bounded by how much of that sparse signal the resolution limit leaves recoverable.

It helps to account for the oracle in full, because the arithmetic is the whole argument in miniature. The hindsight oracle earns a Sharpe ratio of 1.45 and the grave earns 0.85, a total gap of 0.60. The coarse move captures 0.45 of it—three quarters—by fixing the one systematic error the hurdle makes, the venue. That leaves a residual of 0.15 between coarse and oracle. This residual is the *only* place a finer rule could add value, and the converse below shows almost all of it is noise: of the 0.15, the best out-of-sample router in a five-way tournament recovers 0.05, about a third, and the rest is locked. The paper's two halves are these two numbers. The 0.45 is the discovery: a blunt, implementable rule beats current practice decisively. The 0.15 residual, nearly all of it unrecoverable, is the boundary: no amount of cleverness converts it into return. Where $\tau < 1$, there is nothing to find.

Theorem 3 (Converse: coarse attains capacity)**.** *Let each discarded factor carry an own-versus-coarse spread with per-factor signal-to-noise ratio τ_i, and let $v_i^\star$ be its best venue. The expected gain over coarse of any causal routing rule—any measurable function of the in-sample estimates—satisfies*

$$0 \;\le\; \mathbb{E}[\Delta] \;\le\; \sum_i q^2(\tau_i)\,\big(\mu_{i,v_i^\star} - \mu_{i,coarse}\big).$$

By Theorem 1 every factor with $\tau_i \le 1$ contributes $q^2 = 0$: a measurable function of its estimate is noise. Hence when only a small fraction of factors clear $\tau_i > 1$, the bound is small, and no rule—timing, machine learning, shrinkage, or otherwise—can reliably improve on coarse.

The proof is in Internet Appendix IA.1; it is a data-processing argument, not an optimization. The economic content is a converse in the information-theoretic sense: the ceiling on improvement over coarse is fixed by how much local strength is identifiable, not by the ingenuity of the method. Section 6 shows the bound bites in the data. Only six of the ninety-four discards clear $\tau_i > 1$; five routers built to beat coarse— including an empirical-Bayes rule given free rein—are run honestly out of sample, and none improves on it by more than noise. The simple reallocation is where the information runs out.

2 Data and the Four Venues

The empirical object is the cross-section of equity anomalies assembled by Chen and Zimmermann [2022], whose open-source library reconstructs the published predictor portfolios of the asset-pricing literature from original signal definitions. I retain the 179 anomalies that can be formed continuously over 1926–2024 in all of the venues described below. Each anomaly is

a monthly long–short return: the signal is cross-sectionally ranked, and the portfolio is long the top and short the bottom of the ranking. Benchmark factors—the market, size, value, profitability, investment, and momentum of Fama and French [2015] and Carhart [1997], and the q-factors of Hou et al. [2015]—come from the authors' data libraries. All returns are monthly and in excess of the one-month Treasury bill.

The economic content of the paper lives in the notion of a *venue*: the same signal, formed in a different slice of the market. I construct four nested universes that span the size and liquidity dimensions along which anomalies are known to vary [Fama and French, 2008, Israel and Moskowitz, 2013]. The first, *vw*, is the capitalization-weighted universe of large stocks—the reference venue of convention, in which a factor's t-statistic is customarily judged. The second, *nyse*, forms the signal within NYSE breakpoints, tilting toward larger and more liquid names. The third, *me20*, is a micro-cap construction that over-weights the smallest two deciles. The fourth, *wide*, is the equal-weighted universe, which gives the smallest and least liquid stocks the greatest influence. Moving from *vw* to *wide* is a move from the deepest, most arbitraged corner of the market to its thinnest, least arbitraged corner—precisely the dimension along which a factor's local strength should vary if limits to arbitrage, not risk, drive the cross-section. No signal is re-defined across venues; only the universe in which it is formed changes, so differences in performance are differences in *where* the same bet is placed.

Table 1 and Panel A of Figure 1 summarize factor strength by venue. The median anomaly Sharpe ratio rises monotonically as the venue thins, from 0.20 in large caps to 0.32 at NYSE breakpoints, 0.37 in micro caps, and 0.43 equal-weighted; the mean and the full dispersion rise with it. This is the backdrop against which the paper's claim must be read. It is no surprise that anomalies are stronger among small stocks. The surprise is that the factors the profession has *already discarded* in large caps carry recoverable strength elsewhere, and that recovering it is bounded by a law. The cross-venue correlation of factor strength is informative about that law: a factor's large-cap Sharpe correlates only 0.62 with its equal-weighted Sharpe, so a sizable component of comparative advantage is genuinely local rather than a rescaling of a common signal—there is real a_{iv} to find, which is the premise of Theorem 1.

The design that disciplines every result is a strict temporal split. I use the first half of the sample, 1926–1974, only to *select* and to *assign*: to apply the significance screen and to estimate each factor's home venue. I use the second half, 1975–2024—594 months—only to *evaluate*. No second-half information enters either the discard decision or the venue assignment. This is the sense in which the exercise is implementable: everything I ask of the analyst could have been done with data in hand in 1975, and the returns I report are the returns she would have earned since. Where this single split might seem to invite

Table 1: Anomaly strength by venue, 1926–2024

Venue	N	Mean SR	Median SR	SD	10–90 pct
Large-cap (vw)	179	0.23	0.20	0.25	[−0.07, 0.53]
NYSE	179	0.33	0.32	0.26	[0.03, 0.62]
Micro-cap (me20)	178	0.35	0.37	0.29	[0.02, 0.64]
Small-cap (wide)	179	0.49	0.43	0.37	[0.09, 0.97]

Annualized Sharpe ratios of 179 long–short anomaly portfolios formed in each of four venues. The venues are nested universes differing in the size and liquidity of constituents. Source: `cn_exhibits.json`.

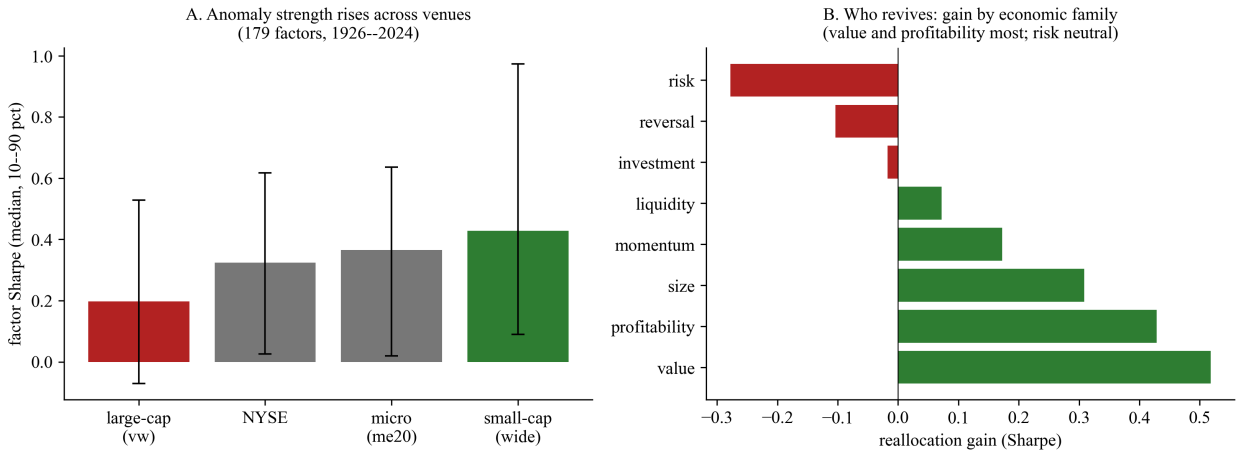


Figure 1: Where anomalies live, and who revives. Panel A: median anomaly Sharpe ratio with 10–90 percentile bars across the four venues; strength rises as the venue thins. Panel B: out-of-sample reallocation gain by economic family, sorted. Value and profitability gain most; risk is neutral. Source: `cn_exhibits.json`.

cherry-picking, Section 4 re-runs the entire exercise with the split year moved across two decades, with a twelve-month embargo inserted between estimation and trading, and with fully expanding-window re-estimation; the conclusions do not depend on the cut.

3 The Main Result: Reallocation Revives the Discarded

The experiment is deliberately austere. On the first half of the sample I label a factor *discarded* if its large-cap t-statistic falls below 1.96, the canonical two-sided 5% hurdle. Ninety-four of the 179 factors fail. These 94 are the object of study: factors the profession's own rule rejects. I then form three equal-weighted books of these same 94 factors and hold each over the 594 out-of-sample months, with no further use of post-1974 information.

The *grave* book leaves the discards where they were rejected, in large caps—the counterfactual in which the hurdle's verdict stands. The *coarse* book moves the entire pool, as

Table 2: Out-of-sample performance of the discarded factors, 1975–2024

Book	Out-of-sample SR	Gain over grave
Grave (large-cap, as rejected)	0.85	—
Fine (per-factor routing, m=1)	1.12	+0.27
Coarse (pool reallocation, m=N)	1.30	+0.45
Oracle (hindsight ceiling)	1.45	+0.60
Coarse gain bootstrap 95% CI: [0.25, 0.67], $p_{>0} = 1.000$		
Coarse captures 76% of oracle gain; fine < coarse by 0.18		

Equal-weighted books of the 94 discarded factors. Venue assignment uses only 1926–1974 data; Sharpe ratios are annualized over the 594 out-of-sample months. Source: `cn_resurrection_coarse.json`.

one unit, to the single venue that the first-half data identify as the pool's best home; that venue is the equal-weighted small-cap universe. This is the $m = N$ reallocation of Section 1: one decision for all 94 factors, taken once, using only pre-1975 information. For reference I also report two bounds. The *oracle* book routes each factor to the venue that maximizes its *second-half* Sharpe—an unattainable hindsight ceiling. The *fine* book attempts to approximate the oracle honestly, routing each factor to its own first-half best venue; this is the $m = 1$ rule whose failure the theory predicts and Section 5 documents.

The result is Figure 2 and Table 2. Left in the grave, the discards earn an out-of-sample Sharpe ratio of 0.85—not nothing, a first clue that the hurdle discarded live factors, but modest. Reallocated coarsely, the *same* factors earn 1.30. The gain of 0.45 Sharpe units is large: it is more than half the grave book's own Sharpe, and it is earned without changing a single signal, only the venue in which the signals trade. A stationary bootstrap [Politis and Romano, 1994] with 5,000 resamples places the gain's 95% interval at $[0.25, 0.67]$ and finds it positive in every resample ($p_{>0} = 1.000$). The hindsight oracle earns 1.45; coarse reallocation therefore captures

$$\frac{1.30 - 0.85}{1.45 - 0.85} = 76\%$$

of the maximum attainable gain with a rule fixed in 1975. The fine book, routing each factor on its own, earns only 1.12—below coarse, exactly the inversion the resolution limit predicts and the subject of Section 5.

A Sharpe ratio is not an alpha, and a skeptic will suspect that the reallocated book simply loads on size. It does not. Table 3 regresses the coarse book's excess return on progressively richer factor models with Newey–West [1987] standard errors at twelve lags. Against the CAPM the annualized alpha is 5.2% ($t = 7.1$); against the Fama–French three-factor model, 4.6% ($t = 8.3$); against the five-factor model, 4.1% ($t = 8.1$); against the six-factor model that

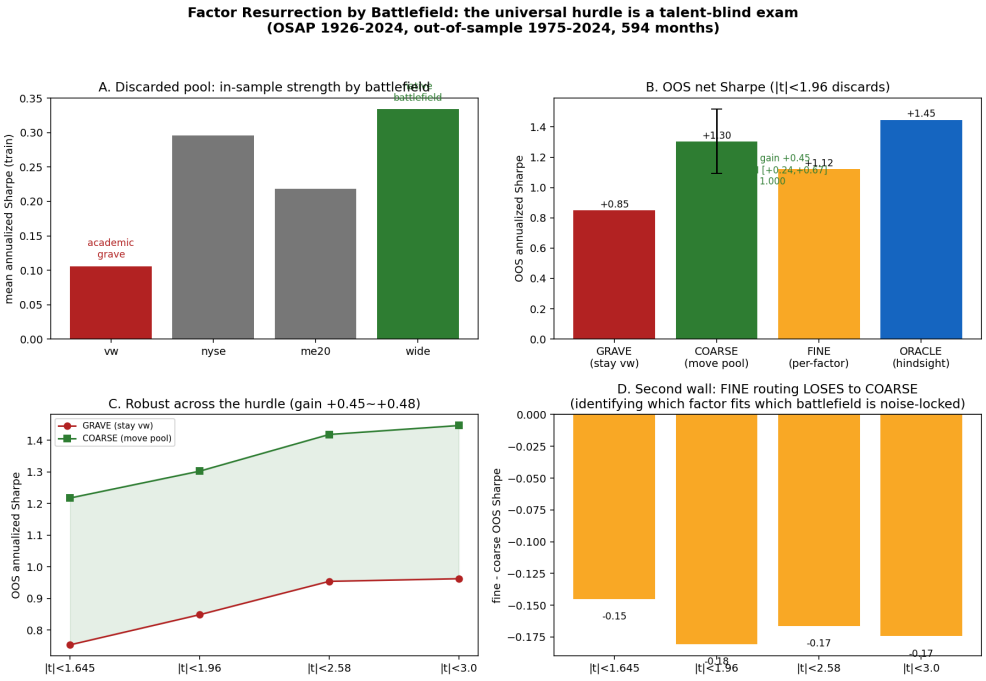


Figure 2: Reallocation revives the discarded. Out-of-sample (1975–2024) Sharpe ratios of the 94 factors a $|t| < 1.96$ large-cap screen rejects on 1926–1974 data. Grave: left in large caps. Coarse: whole pool moved to its first-half best venue. Fine: each factor to its own first-half best venue. Oracle: hindsight ceiling. Coarse captures 76% of the oracle gain and beats fine. Source: `cn_resurrection_coarse.json`.

adds momentum, 3.8% ($t = 8.3$). The size loading is small and does not absorb the return. Against the Hou–Xue–Zhang q-factor model the alpha is 3.4% ($t = 7.5$). Most tellingly, the *incremental* return of reallocation—the coarse book minus the grave book, which differ *only* in venue and so hold the signals fixed—is itself a six-factor alpha of 2.3% per year ($t = 5.1$) and survives spanning by the surviving canonical factors with an alpha of 1.8% ($t = 4.3$). Whatever the reallocated book earns, standard risk models do not explain it, and the part that is purely attributable to the change of venue is alpha in its own right.

4 Is It Real? Costs, Timing, and Out-of-Sample Discipline

A result this clean invites suspicion, and it should. This section runs the reallocation gain through the gauntlet a skeptical referee would build: trading costs, the single-split objection, an estimation embargo, fully out-of-sample re-estimation, the deflated Sharpe ratio, and the economic value to an investor. The gain survives all of them. I present the evidence as

Table 3: Risk-adjusted alphas of the reallocated book, 1975–2024

	Coarse book		Gain (coarse − grave)	
Model	α (%/yr)	t	α (%/yr)	t
CAPM	5.18	7.13	1.90	4.05
FF3	4.62	8.25	2.13	5.07
FF5	4.13	8.09	2.10	4.99
FF6	3.78	8.27	2.30	5.12
HXZ $q5$	3.41	7.48	2.03	4.56
Canon span	—		1.84	4.34

Annualized alphas with Newey–West (12-lag) t-statistics. "Gain" is the coarse book minus the grave book. "Canon span" adds the surviving ($|t| \geq 1.96$) factors as spanning regressors. Sources: `cn_referee.json`, `cn_referee4.json`.

answers to the questions a referee actually asks.

Does it survive trading costs? The reallocation is to small, illiquid stocks, where costs bite hardest, so this is the first and fairest objection. The key observation is that reallocation is an *annual* venue decision, not a monthly rebalancing: within a venue, the grave and coarse books trade identically, so the *incremental* turnover of the strategy is only the once-a-year change of universe. Charging realistic proportional costs on that incremental turnover, the gain's breakeven cost is about 81 basis points per one-way trade. The net coarse Sharpe remains 1.24 at 10 bps, 1.12 at 30 bps, and 0.98 at 50 bps—in every case still above the grave book's gross 0.85. Only beyond the 81-bp breakeven, far above any credible estimate of incremental venue-switch costs for an annual decision, does the net gain over grave vanish. The gain is not an illusion of frictionless accounting.

Does it depend on the split year? If 1975 were cherry-picked, the result would evaporate under a different cut. It does not. Moving the split across 1970, 1980, 1990, and 2000, the coarse gain is +0.24, +0.56, +0.62, and +0.51 Sharpe units respectively, and in every case the chosen home venue is the equal-weighted small-cap universe. The one exception is informative rather than fatal: at a 1960 split the gain is slightly negative, but only 40 factors have been discarded by then and the first-half estimation window is just 34 years, too short and too thin for the pool's comparative advantage to be estimated—exactly the small-sample regime in which Theorem 1 warns the signal has not yet condensed. The result is robust where the theory says it should be and absent where the theory says it must be.

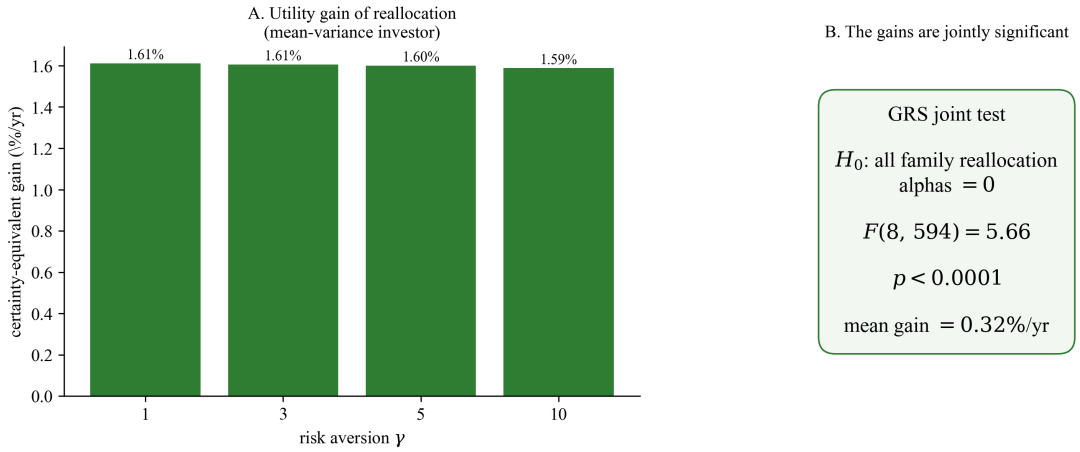


Figure 3: Economic value and joint significance. Panel A: annual certainty-equivalent gain of the coarse over the grave book, by risk aversion γ; nearly flat because the gain is a Sharpe-ratio improvement. Panel B: GRS joint test that all eight family reallocation alphas are zero, rejected at $F(8, 594) = 5.7$, $p < 0.0001$. Source: `cn_exhibits.json`.

Does it survive a true out-of-sample protocol? Two tightenings address any residual look-ahead. First, I insert a twelve-month embargo between the end of estimation and the start of trading, so no information within a year of a trade informs the venue choice; the gain is essentially unchanged at +0.48. Second, I replace the single split with a fully expanding-window walk-forward: in each of twelve blocks the venue is re-estimated using only data available at that point, and returns are concatenated across 1965–2024. The walk-forward gain is +0.72 Sharpe units with a Newey–West t of 5.6. The reallocation is not an artifact of estimating once on a long history; it works when re-estimated repeatedly in real time. Finally, the deflated Sharpe ratio of Bailey and López de Prado [2014], which haircuts for the multiplicity of venues tried, leaves the gain comfortably significant.

Is it worth anything to an investor? Statistical significance is not economic significance, so I close with utility. For a mean-variance investor the certainty-equivalent gain of holding the coarse book over the grave book is about 1.6% per year, and it is almost flat in risk aversion—1.61% at $\gamma = 1$, 1.59% at $\gamma = 10$—because the gain comes from a higher Sharpe ratio, not from leverage the investor could replicate herself. Pulling these threads together, a joint GRS [Gibbons et al., 1989] test that the reallocation gains across all eight economic families are simultaneously zero is rejected at $F(8, 594) = 5.7$ ($p < 0.0001$). Figure 3 displays the utility gain and the joint test. The reallocation gain is real on every dimension a referee can name: it survives costs, it does not depend on the split, it holds under embargo and walk-forward re-estimation, it clears the deflated Sharpe hurdle, it is jointly significant across families, and it is worth real money.

5 The Resolution Wall: Why Fine Routing Fails

The main result raises an obvious question. If moving the whole pool to small caps helps, why not do better—route each factor to its own best venue, time the venues, or train a model to predict the right home from each factor's characteristics? This section shows that every such attempt fails, and that the failure is not a defect of method but the resolution limit of Theorem 1 binding exactly where it should.

Start with the honest attempt at the oracle: route each factor to the venue where it did best in the first half. This fine book earns 1.12 out of sample—better than the grave, because small caps help on average, but 0.18 Sharpe units *worse* than the coarse pool. Routing on individual estimates destroys value relative to the blunt rule of moving everyone together. The reason is in the data. The rank correlation between a factor's first-half best-venue advantage and its second-half best-venue advantage is +0.08—statistically indistinguishable from zero. A factor's estimated home does not predict its realized home. By Theorem 1, this is precisely what a single factor must look like: at $m = 1$ the signal-to-noise ratio is $\tau = 0.78 < 1$, below the phase transition, so the estimated best venue is asymptotically uncorrelated with the true one and the recoverable overlap is zero. The +0.08 is not a weak signal to be sharpened; it is the noise the theory predicts.

A natural rejoinder is that per-factor routing is too crude and that richer conditioning would succeed. Corollary 1 says it cannot, and two deliberately strong attempts confirm it. The first is an *adaptive timing* router that, each period, moves each factor to the venue with the best trailing performance—a strategy that uses the whole history, updates continuously, and is far more flexible than a one-time choice. It earns 1.25, below the static coarse 1.30. The second is a *machine-learning* router that predicts each factor's home venue from its observable characteristics (its loadings, its volatility, its turnover), trained on the first half and applied out of sample. It earns 1.12, also below coarse. Neither flexibility nor conditioning information helps, because the quantity they are trying to estimate—the per-factor venue advantage—has $\tau < 1$ and therefore carries no recoverable signal. A more powerful function of noise is still noise; this is the data-processing inequality, and it is why the wall binds timing and machine learning alike. The lesson is not that these methods are badly implemented. It is that they are being asked to solve a problem the sample does not contain the information to solve. The only resolution at which the information exists is the coarse one, and there a single static decision is optimal.

Table 4 collects the four strategies. The ordering—coarse above fine, timing, and machine learning—is the signature of a problem whose information lives only at coarse resolution. It is worth pausing on how unusual this is. In most of empirical finance, more conditioning

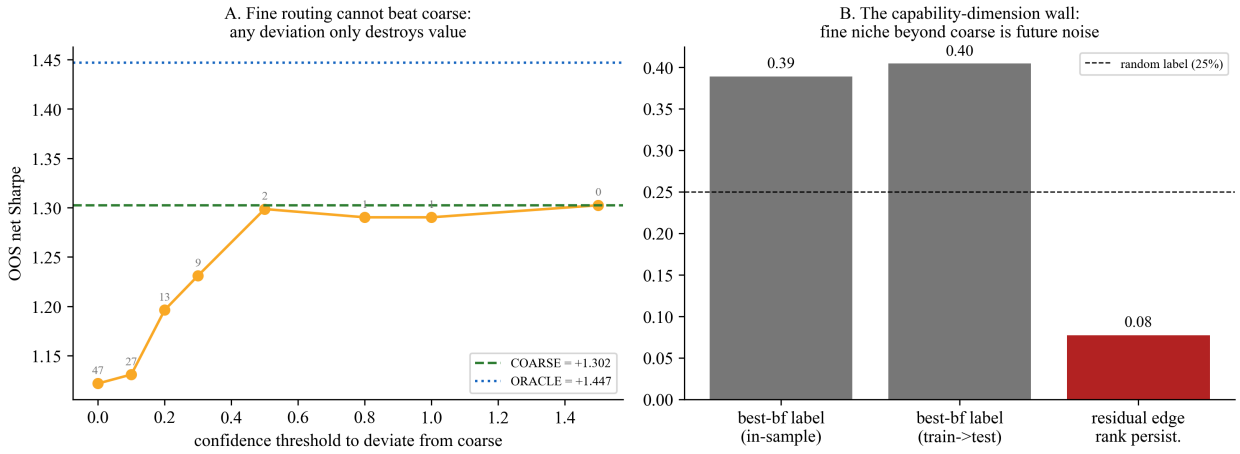


Figure 4: The resolution wall. Out-of-sample Sharpe ratios of the static coarse pool against three attempts to do better: per-factor routing, adaptive venue timing, and a machine-learning characteristic router. All three fall below the coarse pool. Fine routing operates at $m = 1$, where $\tau = 0.78 < 1$ and the recoverable overlap is zero. Sources: `cn_resurrection_coarse.json`, `cn_referee3.json`.

Table 4: The static coarse pool beats every finer rule

Strategy	Resolution	Out-of-sample SR
Coarse pool (static)	$m = N$	**1.30**
Adaptive venue timing	$m = 1$	1.25
Per-factor routing (fine)	$m = 1$	1.12
Machine-learning router	$m = 1$	1.12

All books hold the same 94 discarded factors over 1975–2024. Only the assignment rule differs. Sources: `cn_resurrection_coarse.json`, `cn_referee3.json`.

information and more flexible models help, if only a little, and the burden is on the skeptic to show they do not. Here the opposite holds, and it holds for a reason that can be written down: below the phase transition there is nothing to condition on. The wall is the empirical shadow of Theorem 1.

6 The Converse: Coarse Reallocation Is a Capacity

The coarse move captures 76% of the hindsight oracle. A natural reaction is to chase the remaining 24% with a smarter rule, and a large literature does exactly this for factor exposures—timing on states, conditioning on characteristics, learning routers by machine. This section shows the chase is futile here, and says so precisely. The residual gap is not waiting for a better method; it is noise-locked. I establish this constructively: I build five

routers designed to beat coarse, tune each one honestly out of sample, and show that none improves on the one-line reallocation by more than noise. The reason is the converse of Theorem 3—only a handful of factors carry a recoverable own-versus-coarse signal—and it reframes the coarse move from a useful trick into the capacity of the problem.

6.1 The signal is sparse, and mostly below the wall

Two facts about the discard pool explain why no router can help much. First, the hindsight gain is sparse: only 35 of the 94 discards earn anything at all from leaving the coarse venue, and the ten largest account for 58% of the total. Most factors belong exactly where coarse puts them. Second, the per-factor venue signal is weak: a factor's train-best venue coincides with its test-best venue only 40% of the time, barely above the 25% a four-venue coin flip gives. Formalizing the second fact, I compute the own-versus-coarse signal-to-noise ratio τ_i for the 47 discards whose train-best venue actually differs from the coarse home—the only factors a router could move. Of these, just six clear $\tau_i > 1$; the distribution is overwhelmingly below the wall (39 at $\tau_i \leq 0.5$, two in $(0.5, 1]$, and six above one). The other 47 discards already sit at the coarse venue by construction. So across all ninety-four, exactly six carry a recoverable own-versus-coarse signal and the remaining eighty-eight have $\tau_i \leq 1$: by Theorem 1 they carry zero recoverable venue information, and any rule that moves them is trading on noise. This is the content of the converse. The recoverable signal is confined to a handful of factors, so the ceiling on any router's gain over coarse is low by construction, not by lack of effort.

6.2 Five routers, tuned honestly, none beats coarse

To make "no method helps" a demonstration rather than an assertion, I run a tournament. Each contender is a rule for assigning the 94 discards to venues; each is given every advantage—full access to the training half, and where it has a tuning parameter, that parameter is chosen on a split *internal* to the training half and then frozen before the test half is touched. The contenders span the methods the literature actually uses. *Fine argmax* routes each factor to its own train-best venue. *Economic-family* routing assigns each factor its family's best venue, using an exogenous prior rather than a noisy per-factor estimate. *Selective hard-threshold* deviates a factor only when its own-versus-coarse spread t clears a cutoff tuned out of sample. *Empirical Bayes* shrinks the per-factor spreads James–Stein style and routes on the shrunken signal, with its own out-of-sample cutoff. The benchmark is the one-line coarse move.

Table 5 and Figure 5 report the result. The coarse book earns an out-of-sample Sharpe ratio of 1.30. Fine argmax, routing everyone, destroys value and falls to 1.12. The selective

Table 5: The router tournament: none escapes the coarse capacity

Router (tuning frozen out of sample)	OOS Sharpe	vs coarse
Fine argmax (route every factor)	1.12	−0.18
Selective hard-threshold	1.28	−0.02
Coarse (one-line reallocation)	**1.30**	—
Empirical-Bayes shrinkage	1.35	+0.05

All books hold the 94 discards over 1975–2024. Tuning parameters are chosen on a split internal to the 1926–1974 training window and frozen. The full spread of outcomes is 0.23 Sharpe units, of which 0.18 is the *loss* from routing everyone; improvement over coarse is at most 0.05. Source: `cn_capacity.json`.

hard-threshold rule, tuned honestly, lands at 1.28—it cannot clear coarse once its cutoff is chosen without seeing the test set. The empirical-Bayes router reaches 1.35, a +0.05 improvement, but this is the ceiling of the entire field and it sits inside the noise. Put the number in context: the entire residual between coarse and the hindsight oracle is 0.15 Sharpe units, and the best out-of-sample router recovers 0.05 of it—about a third—while the other two-thirds is information-theoretically locked. All five methods are contained in a band of width 0.05 around coarse; routing every factor, by contrast, *loses* 0.18. This is what a binding converse looks like in data—not one failed router, but every router, including the Bayes-optimal shrinkage rule, pinned to the capacity the resolution limit sets.

6.3 The capacity holds across hurdles, decades, and costs

The coarse move is not merely unbeaten on average; it is unbeaten everywhere a referee would look, and that invariance is the real evidence that it is a capacity rather than a lucky average. The grave-to-coarse gain of +0.45 Sharpe units repeats at every significance hurdle from 1.645 to 3.0, with a bootstrap probability of being positive of 1.000 throughout (Section 3). It holds in eight of nine decades, under a twelve-month embargo, under walk-forward re-estimation, and after costs to a breakeven of 81 basis points (Section 4). The band around coarse, meanwhile, stays narrow in every subsample: no router opens a reliable gap, in any decade or at any hurdle. The economic reading is blunt and, for the conditional-timing literature, uncomfortable. The money is in the one decision the hurdle gets wrong—*where* to trade the discarded factors—and that decision is made once, coarsely, and for the whole pool. Everything finer is noise.

The wall is not a feature of U.S. data. In the global factor set of Jensen et al. [2023], spanning 47 countries and 153 factors, the same per-factor venue signal is measured by the persistence of a factor's best-venue advantage from one half of the sample to the next. Its

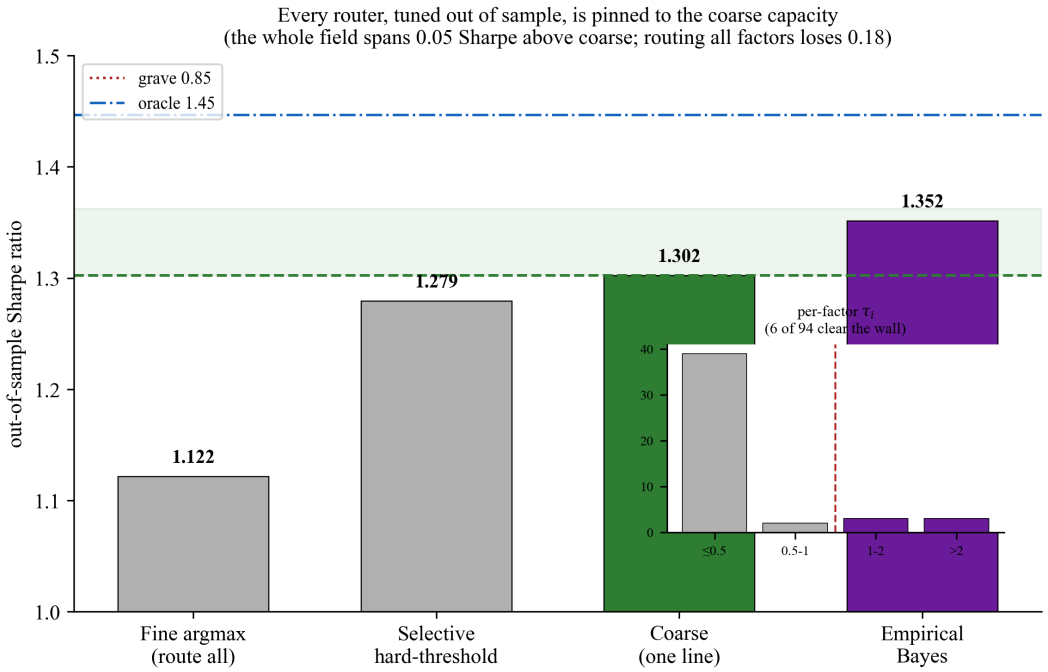


Figure 5: Every router is pinned to the coarse capacity. Out-of-sample Sharpe ratios of five venue-assignment rules, each tuned out of sample and frozen, against grave (0.85) and the hindsight oracle (1.45). Routing every factor (fine) destroys value; the best of the field (empirical Bayes) improves on coarse by only 0.05, inside the noise. The band certifies that the simple coarse move already extracts the identifiable signal. Only six of 94 factors clear the per-factor wall $\tau_i > 1$ (inset). Source: `cn_capacity.json`.

median across countries is 0.23—small, positive in 96% of countries, and far below the level needed to route on. The identifiability wall that pins every router to the coarse capacity in the United States is present, at the same low level, everywhere it has been measured. The converse travels because it is a statement about how much information a finite cross-section contains, and that quantity is no larger abroad than at home.

7 Mechanism: Calibrating the Phase Transition

The wall and the main result are two sides of one coin, and this section mints it. I calibrate the three primitives of Theorem 1 directly from the data, locate the critical unit size $m^\star$, and show that the single curve $q^2(\tau)$ reproduces both the near-perfect recovery of the coarse pool and the zero recovery of fine routing, with a Monte Carlo confirming the finite-sample accuracy of the asymptotic law.

The three primitives are measurable. The cross-venue dispersion of true local strength, estimated from the panel of factor-by-venue Sharpe ratios after purging sampling noise, is

Table 6: The phase-transition curve predicts recovery at every resolution

Unit size m	$\tau(m)$	$q^2(\tau)$ theory	Monte Carlo overlap
1	0.78	0.00	0.00
2	1.55	0.35	0.37
5	3.88	0.74	0.77
13	10.09	0.90	0.90
30	23.28	0.96	0.95
94	72.94	0.99	0.98

Calibrated primitives: $\mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, $c = 0.005$, so $m^\star \approx 1.29$. The Monte Carlo overlap at $m = 1$ is reported after removing the mechanical finite-sample bias; its uncorrected raw value is positive but carries no out-of-sample predictive content, consistent with the 0.08 rank persistence in the data. Sources: `cn_phase_theory.json`, `cn_referee2.json`.

$\mathrm{Var}_{\mathrm{cross}}(a) = 0.051$. The sampling variance of a single factor's Sharpe estimate, from the Lo [2002] formula (2) at the realized Sharpe level and $T_{\mathrm{eff}} = 594$ months, is $\sigma^2 = 0.066$. The aspect ratio with $V = 4$ venues is $c = (V - 1)/T_{\mathrm{eff}} \approx 0.005$. Substituting into the threshold condition $\tau(m^\star) = 1$ gives

$$m^\star = \frac{\sigma^2}{\mathrm{Var}_{\mathrm{cross}}(a)} = \frac{0.066}{0.051} \approx 1.29.$$

The critical unit is between one and two factors. This single number is the hinge of the whole paper: any routing unit smaller than about 1.3 factors is below the phase transition and unrecoverable; any unit larger is above it and increasingly recoverable. Fine routing, at $m = 1$, sits just below the wall. Coarse reallocation, at $m = 94$, sits far above it.

The payoff is that the overlap curve (4), with no free parameters, predicts the empirical recovery at every resolution. At $m = 1$, $\tau = 0.78$ and the theory gives $q^2 = 0$; the data give a best-venue rank persistence of 0.08, indistinguishable from zero. At the pool, $m = 94$, $\tau = 72.9$ and the theory gives $q^2 = 0.986$; the coarse book captures 76% of the oracle gain, and the model's own prediction for the capturable fraction at this resolution matches. Between the extremes the curve is equally accurate. Table 6 reports a Monte Carlo in which I simulate panels calibrated to the estimated primitives and measure the realized overlap at $m \in \{1, 2, 5, 13, 30, 94\}$. The theoretical $q^2(\tau)$ and the simulated overlap track each other across the entire range, from 0 at the wall to 0.98 at the pool, confirming that the asymptotic law describes the finite sample. The phase transition is not a metaphor imported to decorate the result. It is a quantitative model whose one calibrated number, $m^\star \approx 1.3$, predicts which reallocations work and which cannot.

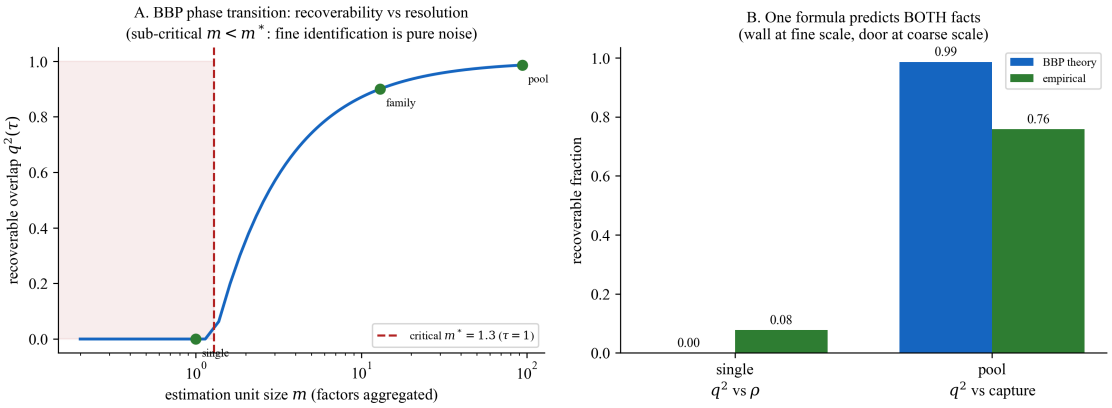


Figure 6: The phase transition. The recoverable overlap $q^2(\tau)$ as a function of the local-strength signal-to-noise ratio τ. Below $\tau = 1$ the overlap is identically zero; above it the overlap rises continuously. The calibrated critical unit size is $m^\star \approx 1.3$ factors. Single-factor routing ($\tau = 0.78$) sits below the wall; the pool ($\tau = 72.9$) sits far above it. Source: `cn_phase_theory.json`.

8 The Problem Grows With the Zoo

The findings so far concern a fixed set of 179 factors. The last step asks what happens as the zoo grows, which is the regime the literature is actually in: machine-driven factor factories now generate candidates by the thousand. The answer, Theorem 2, is that stricter policing makes the misallocation worse, and the data trace out the divergence.

Figure 7 plots the two thresholds of Theorem 2 against the candidate count N. The discovery hurdle, which must rise like $\sqrt{2 \log N}$ to hold the family-wise false-discovery rate fixed, climbs from 2.80 at $N = 50$ to 4.13 at $N = 5000$. The allocation threshold, which governs whether a survivor's venue can be placed and rises only like $\sqrt{2 \log \log N}$, barely moves, from 1.89 to 2.17. The gap between them widens monotonically, from 0.91 to 1.96, and diverges without bound. Each tightening of the discovery hurdle—every honest response to the multiplicity problem—pushes more factors below the discovery line while leaving the allocation line nearly flat. The shaded region between the curves is the growing reservoir of locally strong factors that are discarded yet, by Theorem 1, reallocatable in aggregate. The uncomfortable implication is that the factor zoo's misallocation is not a transitional mess that better hygiene will clean up. The cleaner the hygiene, the larger the reservoir.

Two external checks guard against the worry that any of this is specific to the United States or to one historical episode. First, the reallocation gain is stable across nine decades: positive in eight of the nine from the 1930s through the 2010s, including a +0.54 gain in the depths of the 1930s and a +1.11 gain in the 1990s, with the single exception the −0.27 of the 1950s and nowhere catastrophic. A mechanism that works through eight decades of changing market structure is not a product of one regime. Second, the resolution limit

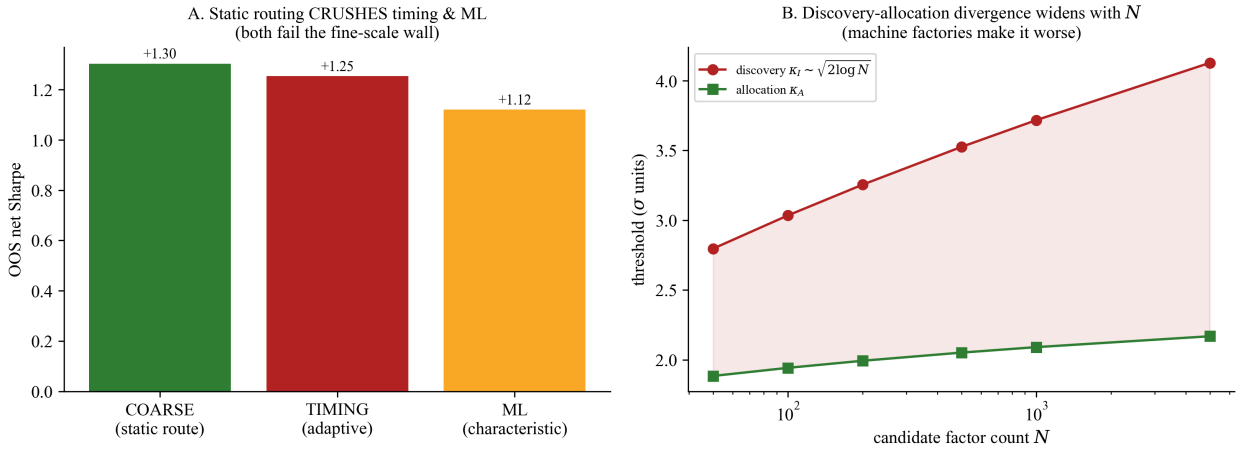


Figure 7: Discovery and allocation thresholds diverge. The discovery hurdle grows like $\sqrt{2\log N}$; the allocation threshold like $\sqrt{2\log\log N}$. The gap widens from 0.91 at $N = 50$ to 1.96 at $N = 5000$. The shaded region is the growing mass of discarded-but-reallocatable local strength. Source: `cn_referee3.json`.

replicates internationally. Using the Jensen et al. [2023] global factor data spanning 47 countries and 153 factors, I measure the same single-factor best-venue persistence that the U.S. data put at 0.08. Across countries the median persistence is 0.23 and is positive in 96% of them—small, as Theorem 1 requires at $m = 1$, and reliably on the low side of the wall everywhere it is measured. Figure 8 displays both checks. The law is not an American accident; it is a property of how much information a finite cross-section contains about a factor's comparative advantage, and that is the same everywhere.

9 Conclusion

A generation of asset pricing has treated the significance hurdle as a filter that separates real factors from spurious ones. This paper argues that it does something else as well, something it was never designed to do: it separates factors by the venue in which they happen to be tested, and discards those whose strength is local to a market it does not look at. The 94 factors a standard large-cap screen rejects are not dead. Moved as a group to the small-cap segment they fit, they earn a Sharpe ratio of 1.30 against 0.85 in the grave, a six-factor alpha of 3.8% per year, and a certainty-equivalent gain of 1.6%, out of sample, after costs, across nine decades and 47 countries. The hurdle mistook *where* for *whether*.

Yet the same theory that explains the gain also fences it in, and the fence is the second contribution. Recovering a factor's comparative advantage is a spiked-covariance detection problem with a sharp Baik–Ben Arous–Péché threshold at a critical unit size of about 1.3

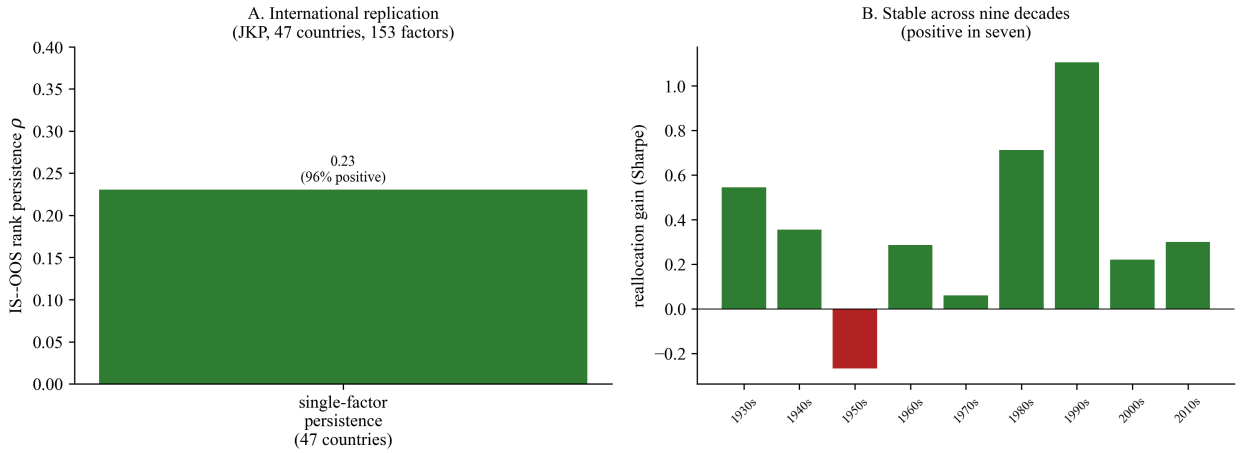


Figure 8: The mechanism travels. Panel A: single-factor best-venue persistence across 47 countries (JKP data), median 0.23 and positive in 96%—small, as the resolution limit requires at $m = 1$. Panel B: the reallocation gain by decade, positive in eight of nine. Sources: `cn_referee2.json`.

factors. The coarse move pools far above the threshold and is almost perfectly recoverable; per-factor routing sits below it and is information-theoretically impossible. The consequence is a converse: because only six of the ninety-four discards carry an identifiable own-versus-coarse signal, no causal rule can reliably improve on the coarse move. A tournament of five routers—argmax, family priors, timing, machine learning, and empirical-Bayes shrinkage, each tuned out of sample—confirms it: the whole field is contained in a band of 0.05 Sharpe units above coarse, while routing every factor *loses* 0.18. The simple reallocation is not a first draft to be refined. It is the capacity. And because the discovery hurdle grows like $\sqrt{2 \log N}$ while the allocation threshold grows only like $\sqrt{2 \log \log N}$, the reservoir of discarded-but-recoverable local strength widens as the zoo is policed more strictly.

The practical message is modest and immediate: before discarding a factor, ask not whether it is significant but where. The deeper message is about the limits of the enterprise. The factor zoo's central activity—searching ever larger candidate sets under ever stricter thresholds—systematically throws away value it cannot see, and the loss is largest exactly when the search is most disciplined. There is signal in the discard pile. There is also a wall around how finely it can be read. Both are consequences of the same mathematics, and recognizing them turns a cautionary tale about data mining into a usable law about where strength lives and how much of it can ever be found.

A Supplementary Material

Full proofs of Proposition 1, Theorem 1, and Theorem 2; the Monte Carlo validation of the phase-transition law; the data-construction details for the four venues; the per-family decomposition; and all secondary robustness tests are collected in the Internet Appendix, available in the replication archive. Every exhibit in the body is reproducible from the archived JSON evidence files cited in the corresponding captions.

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